Everything proposed, on one page: who touches it, what runs it, what it talks to.

Read top to bottom. People, then the agent that operates each service, then the service itself, then the standard interfaces, then the systems the center buys. Built things are solid and warm. Bought things are dashed. Nothing crosses the clinical wall on the right.

Prepared for **Dr. Luis Ramos**, **Fernando Escarzaga**, and **Jesse Martinez**
by **Richard Finlay**, RGF Global Ventures LLC

PEOPLE

- **Rick, operator** — Builds, runs, supervises. Access revocable.
- **Prior auth staff** — Signs every submission and every appeal.
- **Owner and partners** — Ask the owner agent. Approve any sign-off change.
- **Schedulers, front desk** — Scheduling, intake, waitlist. Approve backfill offers.
- **Patients** — Reminders, offers, records requests, inbound calls.
- **Referring physicians** — Orders in, documentation, reports out.
- **Billing staff** — Claims. Their vendor exposes the eligibility feed.
- **Radiologists, readers** — Clinical decisions are theirs alone.

AGENTS

- **Prior auth agent** — Tier B. Reads referrals, flags gaps, drafts appeals. Acts only via the gate.
- **Owner agent** — Tier A. Read-only questions over the center's data. Never acts.
- **Backfill agent** — Tier B. Proposes who gets a freed slot. A human approves.
- **Records agent** — Tier B. Classifies requests, drafts replies. Identity check is human.
- **Voice agent** — Tier C. The vendor's own. Rick specifies, integrates, supervises.
- **No agent. Plain software, no model involved.** — Reminders, intake, report delivery and billing run on the vendors' own products as bought.

HUMAN SIGNS · HUMAN APPROVES · HUMAN VERIFIES

SERVICES

- #1 #2 BUILD — **Prior auth and eligibility engine** — Order detection. Reads the billing vendor's 270/271 feed. Packet assembly, submission, deadline tracking, appeal queue, weekly rates.
- #4 BUILD — **Owner reporting and reader reconciliation** — Joins RIS data to billing data. Monday report, appeal rate as headline. Reader paid-versus-read view.
- #5 #10 BUILD, SMALL — **Backfill and waitlist** — Cancellation in, ranked offer list out. One per-patient rate governor across every sender.
- #6 #7 BUILD, SELECTIVELY — **Records in, prior studies out** — Intake, 30-day clock from receipt, fee rules, DICOM retrieve. Outbound prior-study tracker.
- #9 BUY, I INTEGRATE — **Voice vendor product** — Vendor under its own BAA. Spanish evaluation set. Human escalation on any symptom language.
- #8 #10 BUY, THE RIS — **Reminders, self-scheduling, intake, notices** — RIS native. Verify TCPA caps, Spanish, and events exposed to backfill.
- #12 BUY, RIS AND PACS — **Referrer communication and report delivery** — Standard RIS and PACS function. Referrer volume feeds reporting.
- BUY — **Billing, claims, eligibility feed** — Clearinghouse function. Exposes EDI 270/271. Not rebuilt. The engine reads the feed and acts on it.

THE FDA LINE. NEVER CROSSED.

- #13 DECLINED — ~~Anything touching the image~~ — The FDA line. No image interpretation. No protocol selection. No study recommendation. No appropriateness judgment.

  Software on this side is a device unless it has an FDA pathway. Everything left of the wall stays inside the administrative support exclusion, 21 U.S.C. 360j(o)(1)(A).

Standard interfaces only, secured by four contract terms before signature (FREE): named API access to the center's own data, HL7 interface count and fee schedule fixed, data export rights at no charge, DICOM query permissions.

INTERFACES

- **HL7 v2 ORM, SIU** — EDI 270/271 read; EDI 278; Payer FHIR PA API; Human submission path
- **Data exports** — RIS report export or database; Billing system export
- **HL7 v2 SIU** — Cancellation events; Consent and contact preferences from intake
- **DICOM C-FIND, C-MOVE** — DICOM import; Fax, web form, portal intake
- **RIS scheduling API** — Vendor books against it; Human escalation path
- **RIS native** — Reminder events out to the backfill service; One shared rate cap
- **RIS and PACS native** — Orders in, reports out; Referrer volume to reporting
- **EDI 270/271** — Exposed by the billing vendor; Claims stay with the vendor

- #11 DECLINED, PHASE ONE — ~~Balance-due reminders~~ — Financial content is outside the TCPA exemption. Needs its own consent basis.

SYSTEMS

- **RIS** — Orders, schedule, cancellations, reminders. Chosen by the center.
- **PACS** — Images. DICOM query and retrieve for authorized systems.
- **Billing vendor, clearinghouse** — Claims. Exposes EDI 270/271 and, where supported, 278.
- **Payers and RBMs** — FHIR PA API from 2027-01-01, EDI 278, or a human-submitted packet.
- **Voice vendor** — Own BAA. Telephony and transcripts stored in the US.
- **Prior facilities, El Paso** — Hospitals and imaging groups. Fax or portal out, DICOM back.
- **Referring practices** — Orders and clinical documentation in. Reports out.
- **Model provider** — Named vendor, own BAA, US residency. Replaceable.

- #3 FREE — **Contract terms** — The precondition for every interface here. Free only before signature.

BAA before any PHI flows. 45 CFR 164.502(e). Center, model provider, voice vendor, each under its own agreement.

BAA envelope gap. The AI vendor's agreement excludes Computer Use and Web Fetch, so payer-portal screen-driving against PHI is not available. Design goes through APIs, EDI, and a human step.

Texas SB 1188. Texas patient data stored in the US as of 2026-01-01. Applies to every component on this page, including the model provider.

Two deliverables per build. A deterministic service that runs with no model, plus an agent surface that operates it. The agent is the replaceable part.

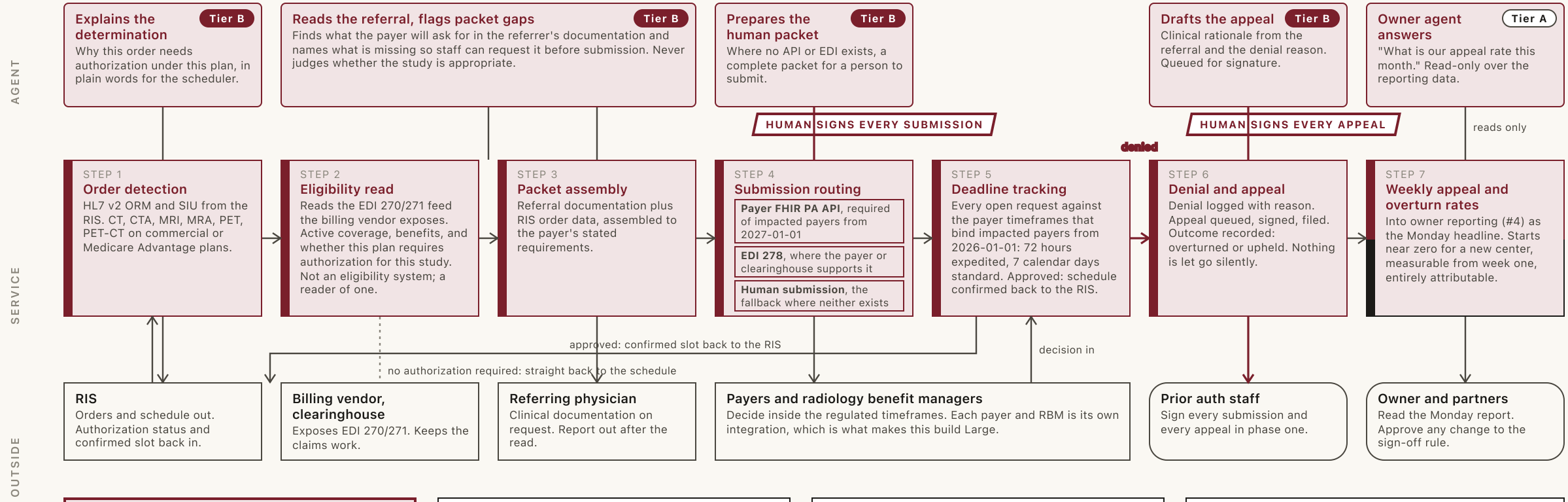
LEGEND — Tier A owner agent, read-only · Tier B workflow agent, acts, a human signs · Tier C patient-facing voice, bought · No agent plain software · Build · Buy · Free · Declined

Rick's engine, licensed to the center with source · Center's outright · HUMAN SIGNS signature gate, every item until the owner approves a change

Prior authorization: one pipeline, three submission paths, a human at every gate.

Left to right is the life of one advanced imaging order. The upper lane is the agent surface. The lower lane is the deterministic service that runs with no model at all. The agent never reaches the world except through a signature.

Candidates **#1** and **#2**
Tier **B**, **Build**, Rick's engine licensed with source



CONSTRAINT, THE MOST FRAGILE ITEM

No portal screen-driving against PHI.

The AI vendor's BAA excludes Computer Use and Web Fetch, the two capabilities one would use to drive a payer portal with no API. So the third path is a human, by design. A locally hosted model is a specification-phase option for that gap. It is not promised.

GRADUATION CRITERION, WRITTEN IN THE SPECIFICATION

Every item, until the owner says otherwise.

Sign-off moves from every item to a sample only after a defined volume at a defined accuracy, measured, and only with the owner approving the change. Until then, every submission and every appeal carries a human signature.

THE FDA LINE, INSIDE THIS PIPELINE

Claims, billing, eligibility. Never appropriateness.

"Claims or billing information" and "determination of health benefit eligibility" are named in the administrative support exclusion, 21 U.S.C. 360j(o)(1)(A). The agent never recommends which study to order or whether a study is clinically appropriate. That would leave the exclusion.

TWO DELIVERABLES

The service is the capability. The agent is the operator.

Steps 1 through 7 run with no model: auditable, testable, in the specification, and what any integrator can take over. The agent lane is the replaceable part. Model vendor, BAA envelope, and pricing can all change without breaking the center.

Dependencies named in the inventory: RIS order and scheduling feed as a contract term; payer FHIR Prior Authorization APIs under CMS-0057-F, which binds payers and not providers; EDI 278 or human submission for plans outside the rule. El Paso payer mix is not verified and nothing here forecasts recovered dollars.

Who touches what, and what each person is allowed to do.

A row is a person. A column is a service. The signature marks are the places the system stops and waits for a human. Radiologists appear once, to show where the system is not.

Access is role-based and logged
Rick's own access is **revocable by the center**

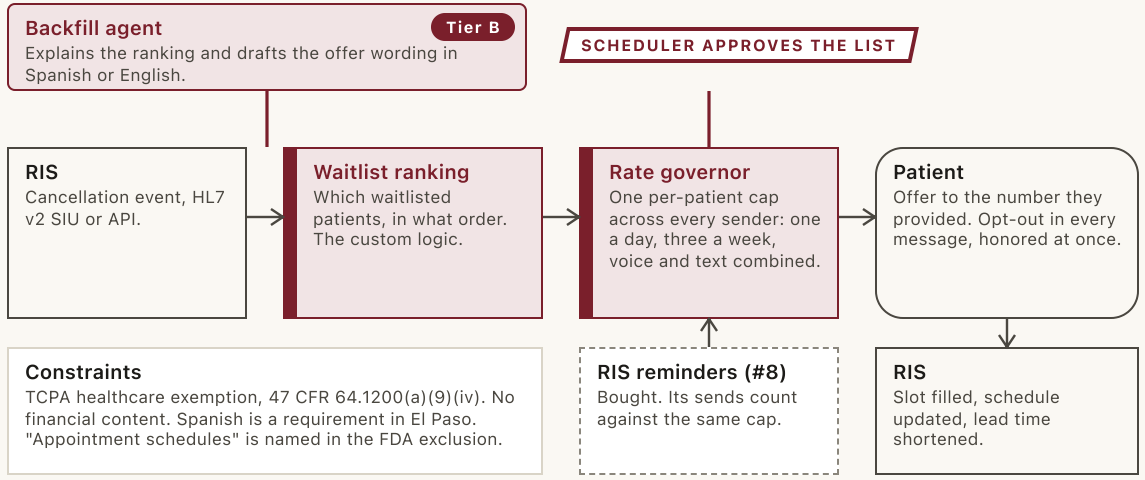
PERSON	PRIOR AUTH AND ELIGIBILITY · TIER B	OWNER REPORTING AND OWNER AGENT · TIER A	BACKFILL AND WAITLIST · TIER B	RECORDS IN, PRIOR STUDIES OUT · TIER B	PATIENT VOICE, VENDOR · TIER C	RIS REMINDERS, SELF-SCHEDULING, INTAKE · NO AGENT
Owner and partners Dr. Ramos, Mr. Escarzaga, Mr. Martinez	See every open request, deadline, denial, and appeal outcome. Approve the graduation criterion that changes the sign-off rule. APPROVES GRADUATION	Read the Monday report. Ask the owner agent open questions: volumes, cancellations, no-shows, payer mix, appeal rate, reader turnaround. Define what the one page contains.	See fill rate and the ranking rules. Change the ranking policy.	See the request log, open clocks, and fees charged.	Hear call recordings and transcripts. Set the escalation rules. Approve the Spanish evaluation bar.	Configure through the RIS vendor. Confirm caps and Spanish at selection.
Prior auth staff The signature	Review every assembled packet and every drafted appeal. Submit by hand on the human path. Nothing goes to a payer without their signature. SIGNS EVERY SUBMISSION SIGNS EVERY APPEAL	Read-only on their own queue metrics.	No access.	No access.	No access.	No access.
Schedulers, front desk	See authorization status on each order so a scan is not booked before it is cleared. Request missing documentation from the referrer.	No access.	Receive the ranked offer list on a cancellation and approve it before anyone is contacted. Maintain consent and contact preferences at intake. APPROVES THE OFFER LIST	Log a walk-in or phoned request so the 30-day clock starts at receipt.	Take escalated calls, including any call with symptom language. Book what the vendor could not.	Operate the RIS. Intake forms and reminders run as the vendor ships them.
Billing staff	See authorization numbers attached to orders for claims. The eligibility feed is their vendor's; the engine only reads it.	Provide the billing export the reporting joins depend on.	No access.	Set the fee schedule inside the permitted components. Retrieval labor is never charged.	No access.	No access.
Radiologists, readers Including Dr. Ramos as a 1099 reader	Untouched. The system never recommends a study or judges appropriateness.	Reader reconciliation view: what was read against what was paid.	Untouched.	Ask for a prior study; the tracker chases it and imports what returns. The read is theirs.	Untouched.	Untouched.
Referring physicians	Orders arrive from them through the RIS. Documentation requests go to them. They are never asked to change an order.	No access. Their volume appears in the owner's view.	No access.	Can request records for their patient; same intake, same clock.	No access.	Receive reports through the RIS and PACS as bought.
Patients	No direct surface. They experience a faster clearance, nothing else.	No access.	Receive an offer for a freed slot, only to the number they gave, in Spanish or English, with opt-out in every message, inside one message a day and three a week across every sender.	Request their own records. Identity verified by a person before anything is released. HUMAN VERIFIES IDENTITY	Call in. Scheduling, directions, prep, status. Told at the start that it is an automated system. Handed to a person on any symptom language.	Reminders, self-scheduling, intake forms, results-ready notices from the RIS.
Rick, operator RGF Global Ventures	Builds and operates the four built services and their agent surfaces. Sees logs and metrics. Cannot sign a submission, approve an offer list, or release a record; those are the center's people. Access granted and revocable by the center; repository under the center's organization.				Specifies, evaluates, integrates, supervises. The contract is in the center's name.	Advises at vendor selection. No operational role.

What the human marks mean. Where a service acts on the world, a named person at the center signs first. That is design, not cost, and it is also the FDA boundary: no software here makes or recommends a clinical decision.

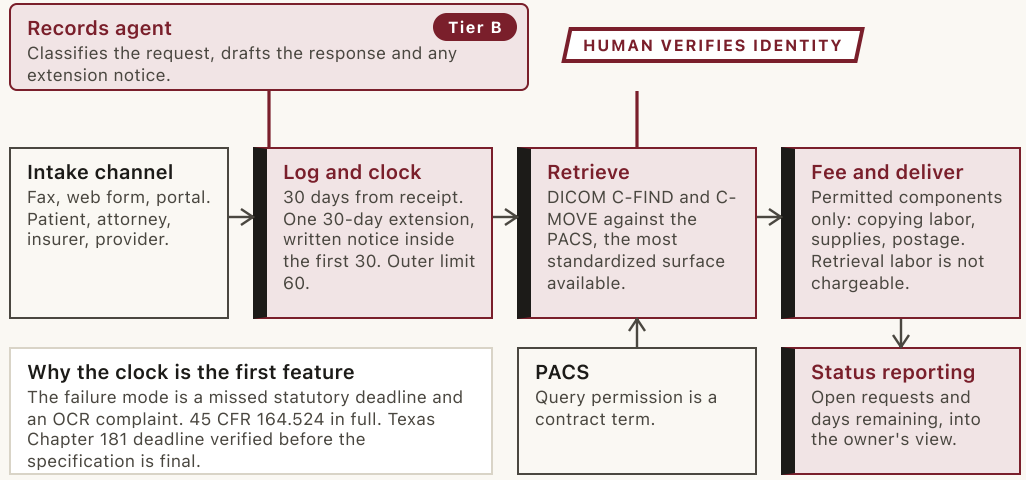
Backfill, records, prior studies, owner reporting: each is a small deterministic service with a bounded agent on top.

Each panel reads left to right
Human gates drawn where they exist

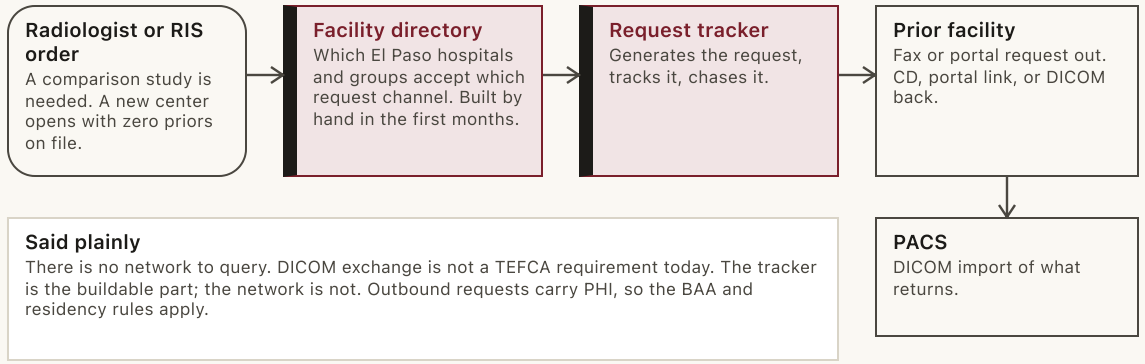
#5 AND #10. BACKFILL AND WAITLIST. TIER B. BUILD, SMALL. RICK'S ENGINE.



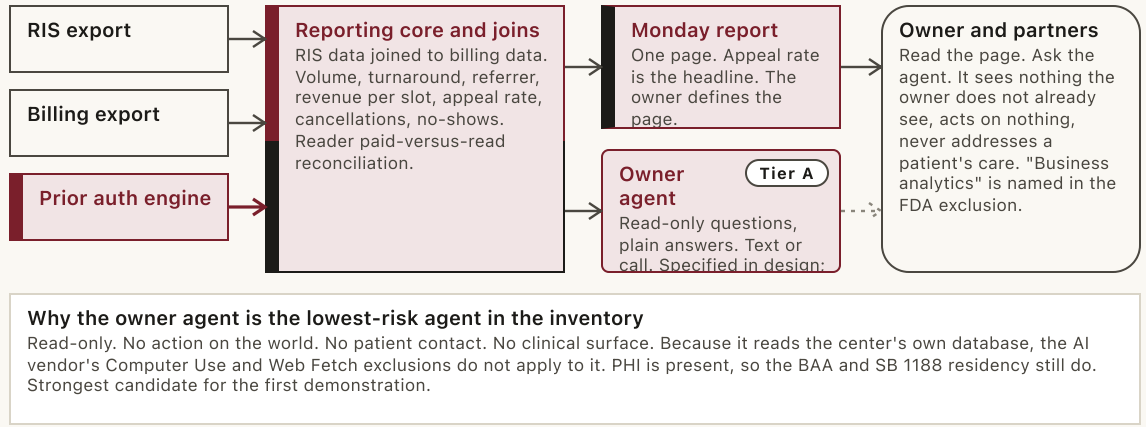
#6. INBOUND RECORDS REQUESTS. TIER B. BUILD, SELECTIVELY. CENTER'S CODE.



#7. PRIOR-STUDY RETRIEVAL. NO AGENT. BUILD, SELECTIVELY. CENTER'S CODE.



#4. OWNER REPORTING. TIER A. BUILD. RICK'S CORE, THE CENTER'S REPORTS.



Every panel follows the same standard: the deterministic service is the specification and what any integrator can take over; the agent is the replaceable operator. Regulatory figures are the regulation's own. No cost or volume benchmark is quoted because none was verified.

Who owns each layer, and the lines the system does not cross.

The left half is a commercial boundary. The right half is a regulatory one. Both are drawn so an engineer who has never met Rick can read them.

Ownership decided **in three tiers**
Boundaries drawn from the statute and the vendor's BAA

OWNERSHIP OVERLAY. MATCH THE STRIPE COLORS TO THE SERVICE BOXES ON PAGES 1 TO 4.

TIER 1

The specification. The center's, outright.

Written for an engineer who has never met Rick. Covers every service on page 1: order detection, packet assembly, deadline tracking, waitlist ranking, the records clock, the reporting joins, and the vendor evaluation criteria for the RIS, PACS, billing, and voice selections. If the partners open a second or third site, the specification is written once and deployed again, one instance per site (page 6).

TIER 2

Center-specific code. The center's, outright.

Configuration, workflows, integrations, data mappings, and reports. In a repository under the center's own organization, with Rick's access revocable. On the diagrams, this is the black stripe:

- Records intake, clock, fee rules, status reporting (#6)
- Prior-study tracker and the El Paso facility directory (#7)
- The Monday report and every center-defined view (#4)
- Every HL7, DICOM, and payer mapping, and every RIS and billing configuration

TIER 3

Rick's existing product, extended for the center. Rick's, licensed.

Licensed to the center perpetually, irrevocably, and transferably, with source, so it can run and modify the code for its own operations indefinitely. The center cannot resell it. On the diagrams, this is the red stripe:

- The prior authorization engine (#1, #2)
- The backfill engine (#5)
- The reporting core (#4)

Boxes with a split stripe hold Rick's engine beneath the center's configuration and mappings.

NOT OWNED BY ANYONE HERE

The RIS, PACS, billing, and voice vendor contracts are in the center's name, never routed through Rick. The model provider is a named vendor under its own BAA, chosen so it can be replaced without touching the services.

COMPLIANCE BOUNDARIES. DRAWN AS WALLS, NOT FOOTNOTES.

THE FDA LINE

Nothing interprets an image, selects a protocol, or recommends a study.

Administrative support software is excluded from the device definition at 21 U.S.C. 360j(o)(1)(A): appointment schedules, claims or billing information, business analytics, practice and inventory management, determination of health benefit eligibility. Every service here lives inside that list. The moment software touches image data or clinical appropriateness it leaves the exclusion and enters Clinical Decision Support territory at 360j(o)(1)(E). Candidate #13 is declined for that reason, out loud.

THE BAA ENVELOPE

What is inside, what is not.

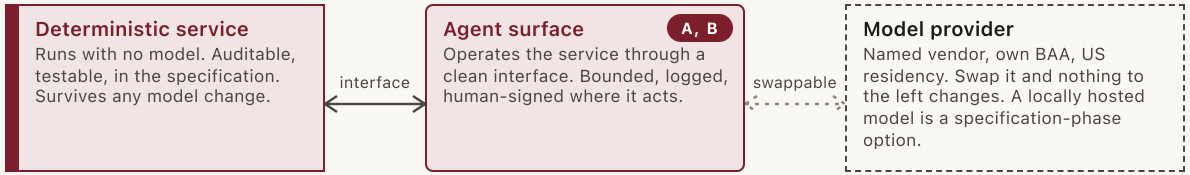
A business associate agreement with the center before any PHI flows, 45 CFR 164.502(e). The AI vendor's agreement covers Skills, Web Search, Memory, Bash tool, and Text Editor tool. It **excludes Computer Use and Web Fetch**, so payer-portal screen-driving against PHI is not available. Confirm the applicable BAA version in writing.

TEXAS SB 1188

US data residency, every component.

Electronic health records with Texas patient data must be physically stored in the US as of 2026-01-01. This binds the RIS, the PACS, the billing vendor, the voice vendor's telephony and transcripts, and the model provider. Verify each one at selection. Texas HB 300 likely reaches Rick directly as a covered entity; its exact intervals are not quoted here because they were not verified.

THE AGENT IS THE REPLACEABLE PART



WHY THIS SHAPE KEEPS THE CONTINUITY PROMISE HONEST

If Rick is gone, the center holds the specification, the center-specific code, a perpetual source license to the engines, runbooks for every procedure, and vendor contracts in its own name. The services keep running with no agent at all. An integrator who has never met Rick can operate them from the specification, and can put a different agent, or no agent, on top. The one thing not promised anywhere in this set is payer-portal automation, which is the most fragile item in the inventory.

Three centers, three instances. The specification is shared. Nothing else is.

A recommendation, not a settled fact. Whether the centers share a back office and a vendor stack is the partners' decision, and either answer fits this page. What does not change is that each center's patient data stays its own, and that this costs more than one shared stack.

Every service on pages 1 to 4 deployed **once per center**, data never pooled

TIER 1, WRITTEN ONCE

One specification, three deployments.

Order detection, packet assembly, deadline tracking, the records clock, the reporting joins, the report assistance path. Written once, with each center's mappings and configuration filled in per site.

FIRST CENTER

Its own instance.

- Its own copy of every service: prior auth, reporting, backfill, records, report assistance.
- Its own database, audit log, and credentials, rotated and reviewed for this center alone.
- Its own model account under its own BAA. US residency verified for this instance.
- Its own RIS, PACS, billing, and voice contracts, or shared ones; the four access terms go into whichever contract applies.
- Its own principals approve its own sign-off rules.

NO DATA CROSSES THIS LINE

SECOND CENTER

Its own instance.

- Its own copy of every service: prior auth, reporting, backfill, records, report assistance.
- Its own database, audit log, and credentials, rotated and reviewed for this center alone.
- Its own model account under its own BAA. US residency verified for this instance.
- Its own RIS, PACS, billing, and voice contracts, or shared ones; the four access terms go into whichever contract applies.
- Its own principals approve its own sign-off rules.

NO DATA CROSSES THIS LINE

THIRD CENTER

Its own instance.

- Its own copy of every service: prior auth, reporting, backfill, records, report assistance.
- Its own database, audit log, and credentials, rotated and reviewed for this center alone.
- Its own model account under its own BAA. US residency verified for this instance.
- Its own RIS, PACS, billing, and voice contracts, or shared ones; the four access terms go into whichever contract applies.
- Its own principals approve its own sign-off rules.

WHY INDEPENDENCE. GOVERNANCE, NOT COST.

It costs more, and that is stated here on purpose.

Three hosts, three databases, three sets of credentials to rotate, three audit logs to review, and the recurring infrastructure line sits in each center's budget rather than once. The reason to pay it: separate principals, separate books, and a line a regulator, an auditor, or a partner's lawyer can see without reading code. In a shared stack, every center's patient data is one breach, one subpoena, and one dispute away from the others.

IF THE PARTNERS CHOOSE A SHARED BACK OFFICE

Share the staff, not the data.

One team can serve three centers: three logins, every session scoped to one center, every action logged against that center. Any report that spans centers is a later, explicit decision by all the principals, built as a read-only view over three databases, never as a merged one. Nothing on this page has to be rebuilt for either answer; what changes is how many vendor contracts carry the four access terms.

WHAT IS SHARED, EXACTLY

The specification and the engines. Nothing that holds a patient.

- The specification, Tier 1, written once.
- Rick's engines, Tier 3: the same code deployed three times, licensed to each center with source.
- The runbooks and the vendor evaluation criteria.
- No database, no credential, no audit log, no model account.

The build starts around March so prior authorization works before your new center opens.

Target opening around June 2027, April at the earliest. Every date here is a target that holds only while the dependencies in the right-hand panel hold. If one slips, the dates move with it.

Added 2026-10-22
after the rehearsal with **Dr. Ramos**

SEQUENCE. DESIGN FIRST, THEN THE BUILD, THEN HYPERCARE.

D1. On signing
Contract language for the four access terms, ready for every vendor.

D2. ~Dec 2026
RIS, PACS, billing, voice evaluation. Working prior auth demo on synthetic data. Owner's gate follows.

D3, D4. ~Jan to Feb 2027
Specifications: prior auth, backfill, records, owner reporting. Payer-specific design once payer mix is known.

B1, B2. ~Mar to Apr 2027
Prior auth agent in the RIS vendor's sandbox, then against payer FHIR and EDI test environments.

B3, B4. ~May 2027 to opening
Staff test it on their own workflows. Live before the doors open.

B5. Opening + 30 days
Hypercare. Then optional support and maintenance, monthly or per patient.

OPERATIONS AGENT

Runs one workflow inside the center's systems.

Prior authorization, cancellation backfill, records requests. Works through the RIS (HL7 v2, DICOM) and the payers' own connections (FHIR prior authorization API, EDI), never by driving a portal screen. Assembles, checks, tracks, drafts. **A person on the center's staff approves anything that leaves the building.**

- Worked example: order arrives in the RIS, coverage and authorization rule checked, request assembled and checked against payer rules, staff approves, submitted electronically, decision clock tracked, approval scheduled, denial drafted as an appeal for a person to sign.

OWNER'S AGENT

Answers the owner, by text or phone. Reads, never acts.

Plain questions over the center's own operational data, answered from the reporting service, not from memory. A Monday report: volumes by modality, reader, and referrer; authorization and denial rates; reconciliation.

- **Calculated and checked:** totals and rates are computed by code and verified; if a figure cannot be verified the agent says so.
- Inside the FDA administrative exclusion: business analytics and practice management by name.
- No patient contact and no clinical surface.

DEPENDENCIES. THE DATES HOLD ONLY WHILE THESE HOLD.

Six things have to be true before the build can start.

- A signed agreement.
- RIS and PACS vendors selected, with sandbox access to their API, HL7, and DICOM interfaces.
- Test access to payer prior authorization connections, FHIR and EDI.
- HIPAA business associate agreements executed, including the model vendor's.
- One named owner contact who approves.
- Expected volumes and payer mix, before payer-specific design starts.

VENDOR-NEUTRAL BY DESIGN

Works with whatever RIS and PACS the center picks.

If the PACS or RIS vendor ships its own version of a feature, the center can use it. The four access terms (API access, HL7 interface count and fees, export rights, DICOM query permissions) keep the center's data reachable whichever way that goes. Guardrails throughout: a person signs anything clinical or outbound; HIPAA BAAs; patient data stays in the United States.